

Caméléon Creator Tutorials
Chapter 3
Manage sub processes

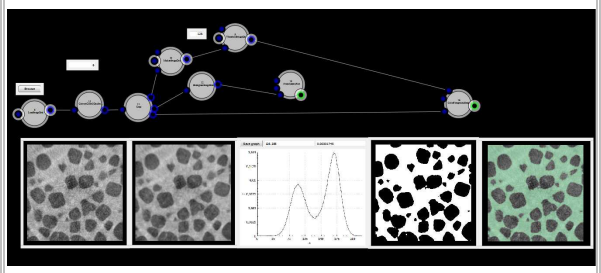
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Caméléon Creator Tutorials
Chapter 3
Advanced prototyping

By Olivier Cugnon de Sévricourt

In this tutorial, we speak about how to use Caméléon to create, from the previous prototype, an advance prototype with sub processes.

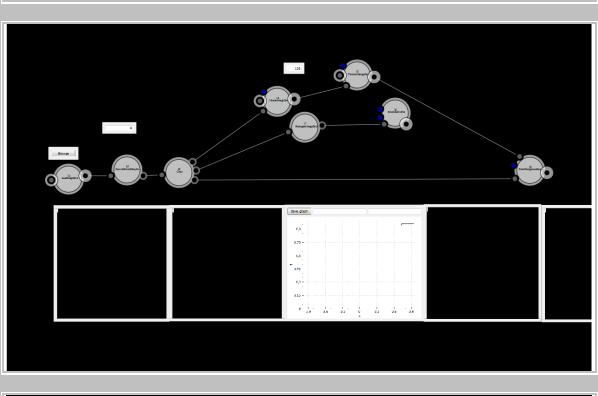
You should follow the previous tutorial.



Open *Threshold-Segmentation.cm.*

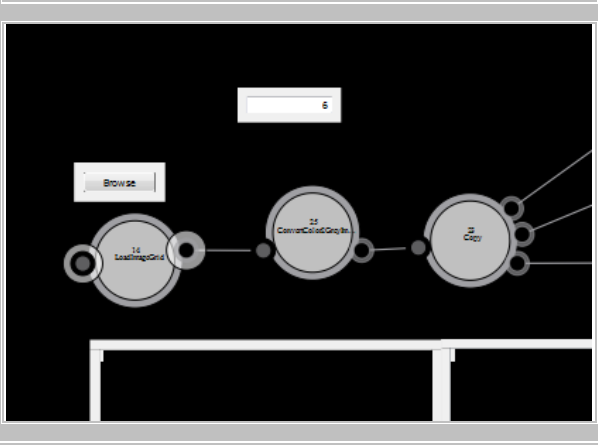
You will need to have this project unmodified in future tutorials.

So, **save** this project as *Threshold-Segmentation-advanced.cm.*



The goal is to simplify this process introducing sub process feature.

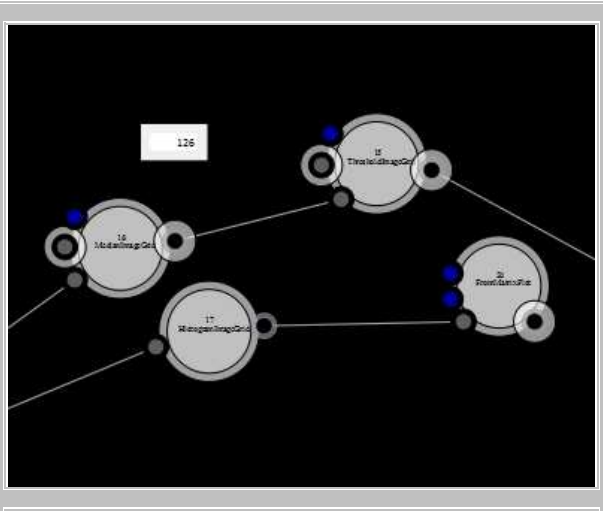
Each sub process must have a user functional scale.



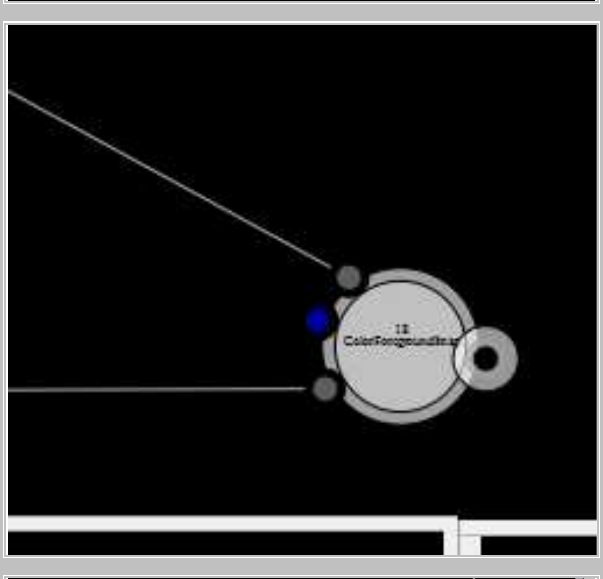
Start by reading our process.

The first 3 operators enable us to load an image.

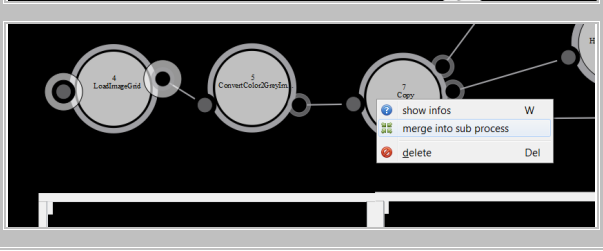
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The second part is like our image processing kernel.

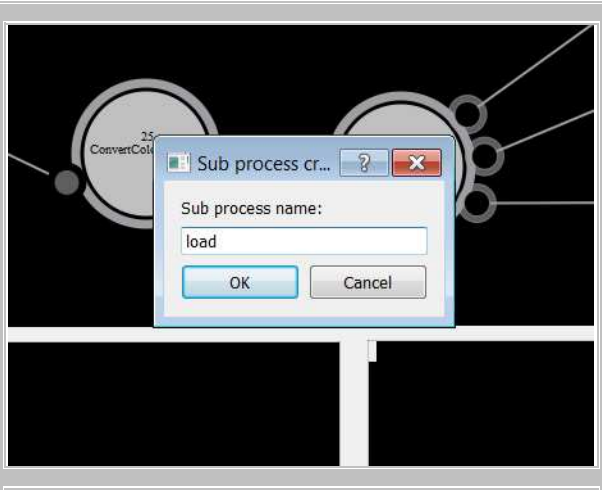
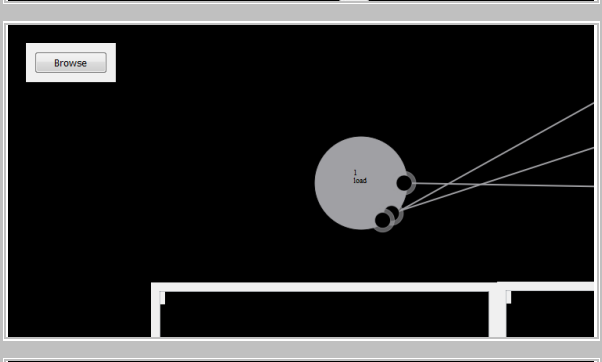
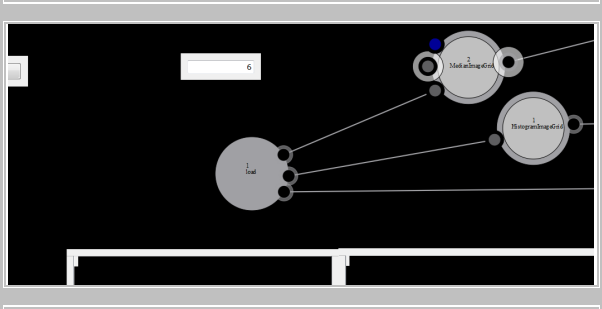
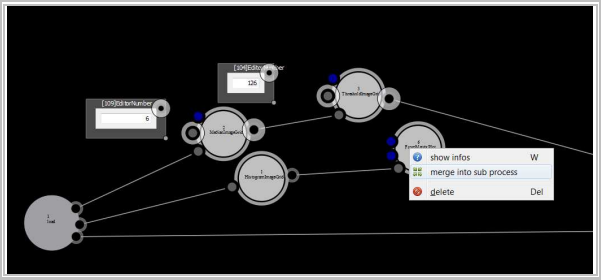


The third part is our save process.

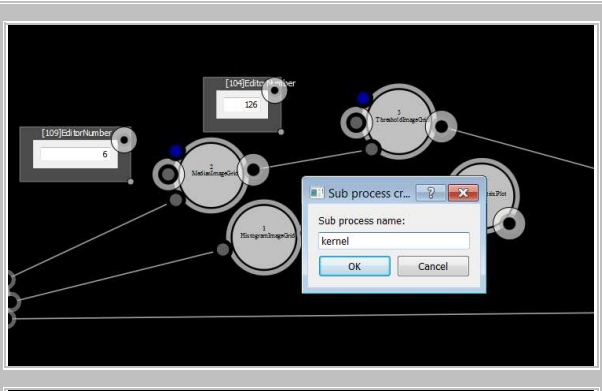


Select the 3 first operators and click left, then **click** on "merge into sub process" button.

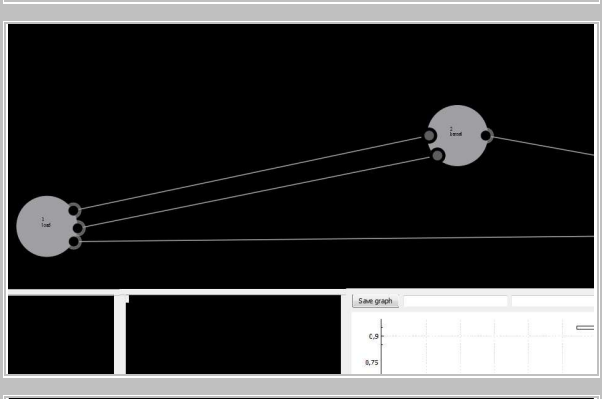
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	A dialog appear. **Set** the name to "load" and **click** on "ok" button.
	You have created your first sub process.
	Like on operator you can **move** connectors to find a good position.
	**Select** our kernel.

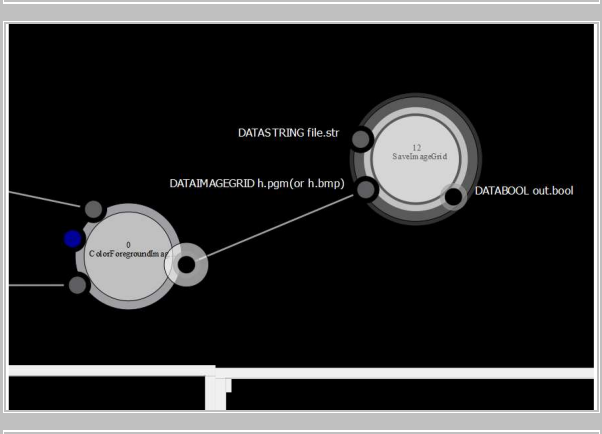
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Create a sub process named "kernel".



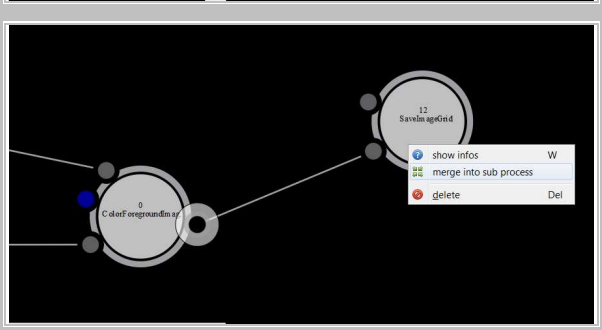
Like this.



We have to finalize our save process.

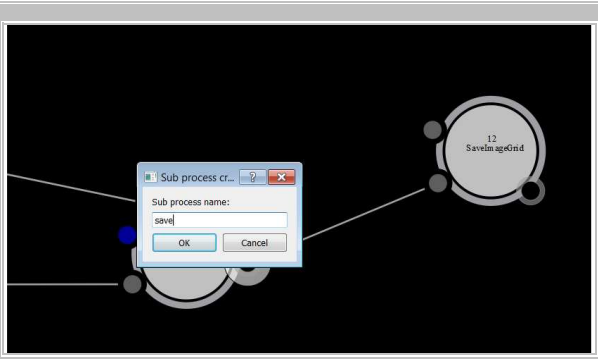
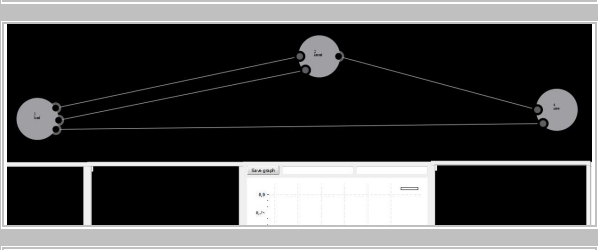
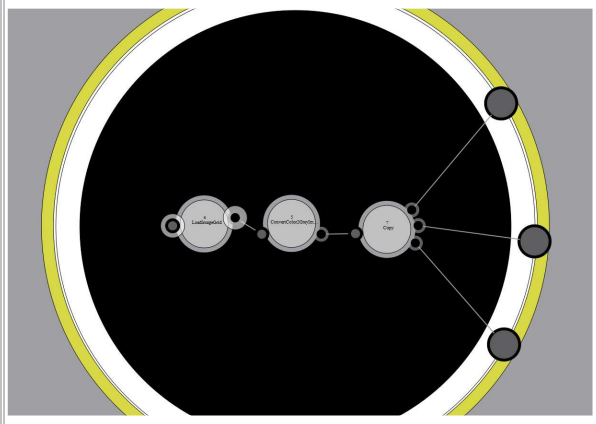
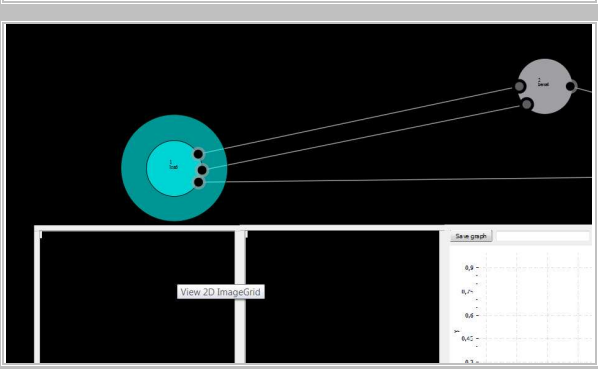
Find & **create** a Save ImageGrid operator.

Connect ColourForgroundImage to Save ImageGrid.

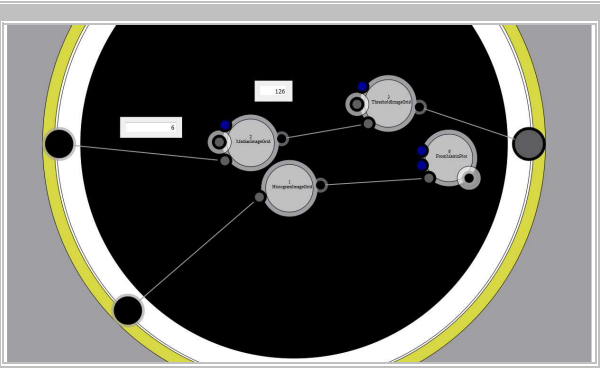
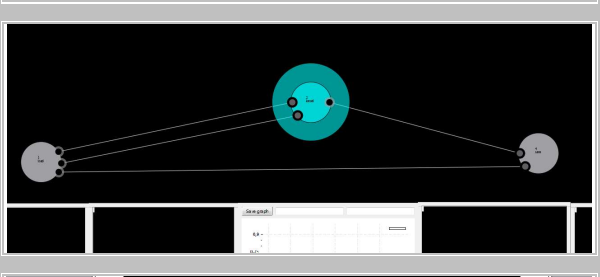
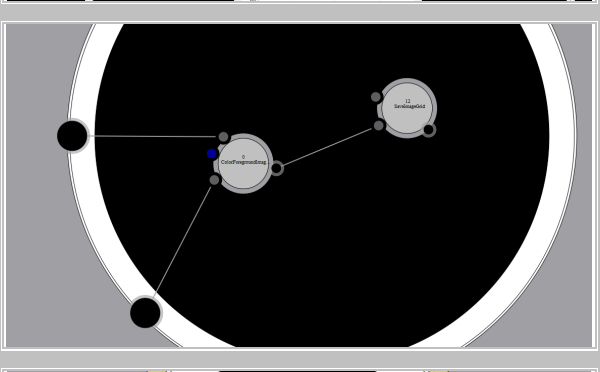
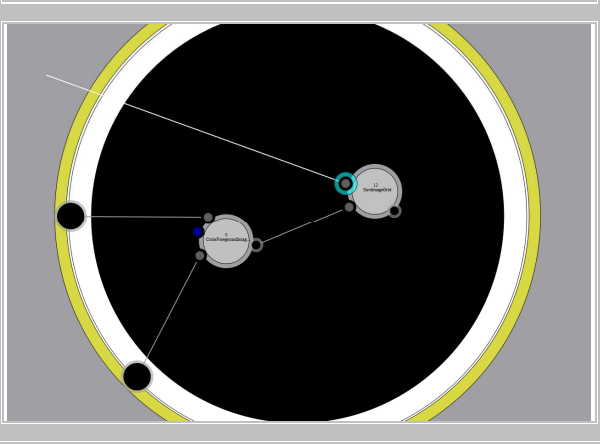


Merge them into a sub process.

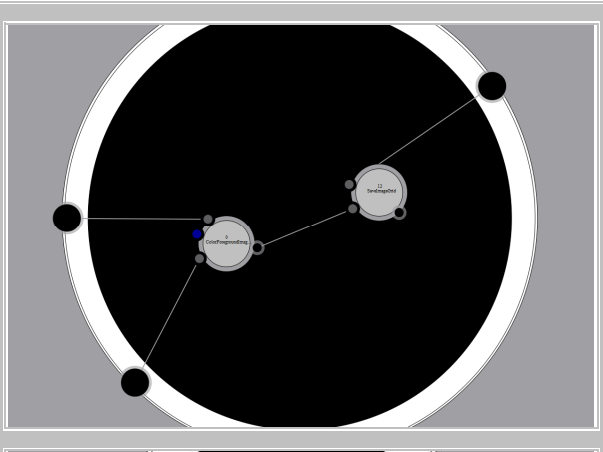
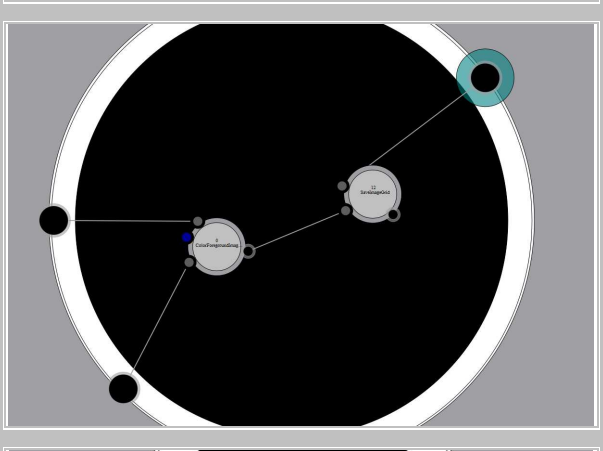
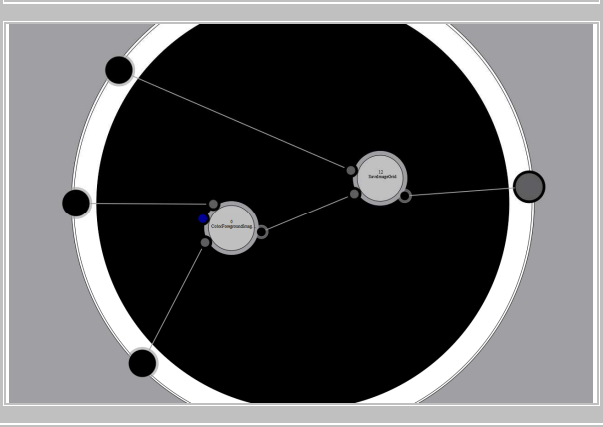
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	**Name** the sub process "save".
	Now, your process look like this. More readable isn't it?
	**Double click** on the load sub process to go into it. Now your in the sub process. Regarding to the top process, you are here in a down scale. Sub process connectors are like bridge from one scale to an other. **Double click** on the sub process black background to go on the upper scale.
	Like this. The sub process you come from is highlighted. **Double click** on kernel to go into it.

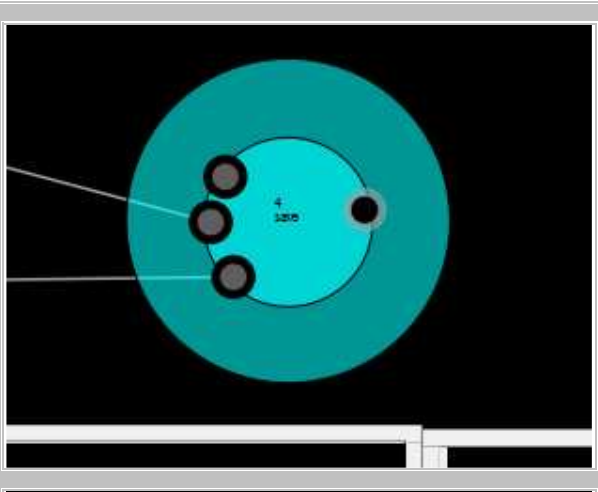
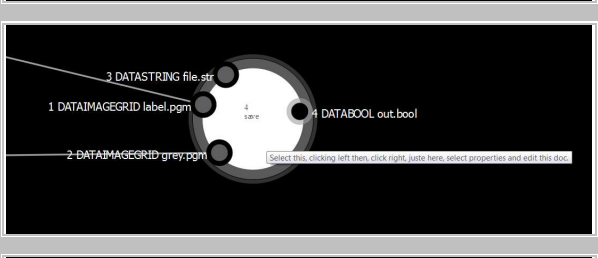
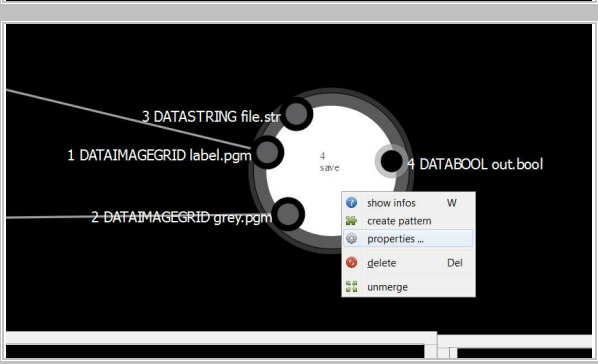
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	**Double click** on the sub process black background to go on the upper scale.
	Like this.
	Go into save sub process. We want to expose to the upper scale an input parameters: the output file path where we will store our result.
	To do this, we **drag** the connector from Save ImageGrid & **drop** it into the grey background. It create a connection between the operator & the sub process.

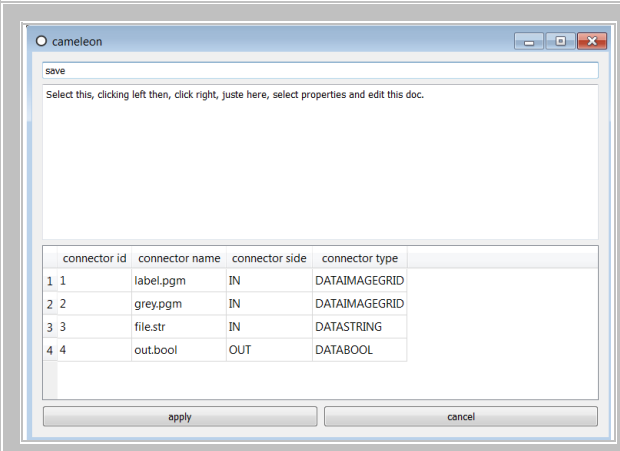
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	We name this action "expose a connector". Now we want to move this connector at a better place.
	We can **move** the sub process connector like we do with operator connectors.
	**Expose** the output connector of Save ImageGrid operator.

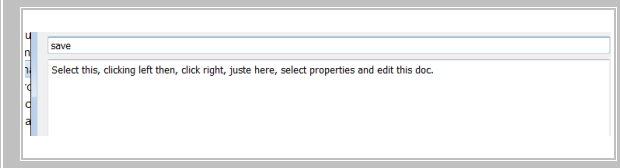
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	**Go** to upper scale.
	**Select** the save sub process. **Press** the W key on the keyboard.
	You see output & input information. By default the name is the same as the inner operator connector connected to the bridge.
	Now, we want to document our sub process. **Click left** on the sub process and **press** on the "properties ..." button.

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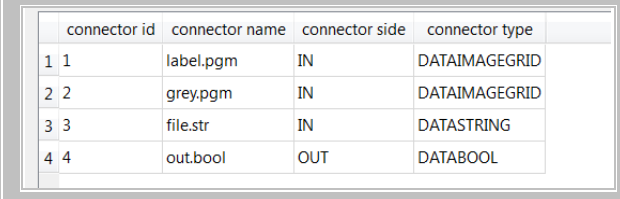


This panel allows you to set your sub process information.



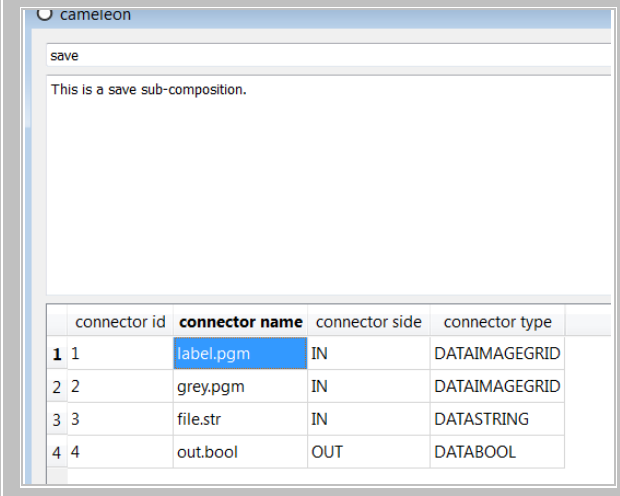
You can **set** its name & its description. This description is shown on the hover tooltip.

We **write** "This is a save sub process".

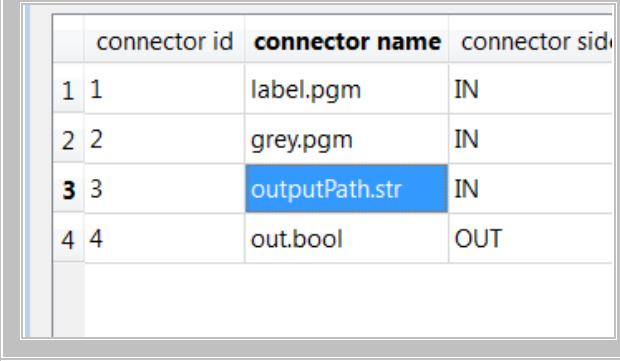


	connector id	connector name	connector side	connector type
1	1	label.pgm	IN	DATAIMAGEGRID
2	2	grey.pgm	IN	DATAIMAGEGRID
3	3	file.str	IN	DATASTRING
4	4	out.bool	OUT	DATABOOL

You have also several connectors information.



You can **edit** connector name by **double clicking** on the connector name.

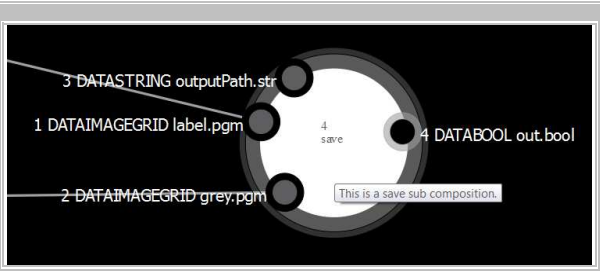


	connector id	connector name	connector sid
1	1	label.pgm	IN
2	2	grey.pgm	IN
3	3	outputPath.str	IN
4	4	out.bool	OUT

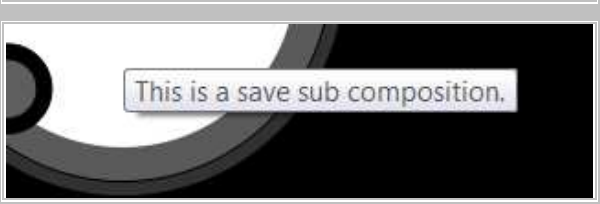
Rename file.str to outputPath.str and **press enter**.

Click on the apply button.

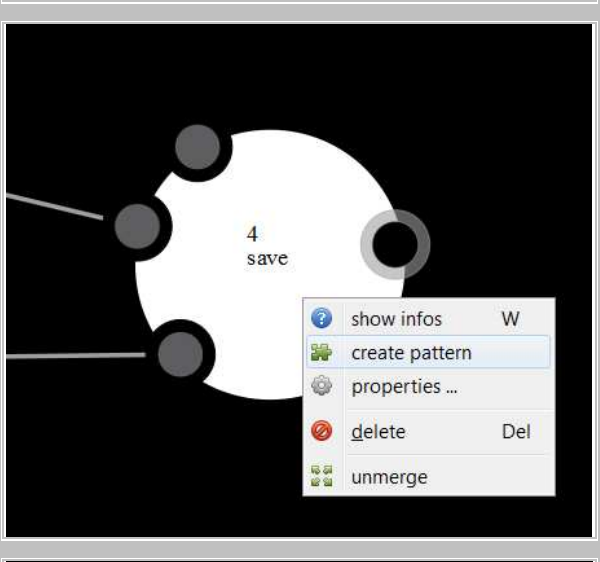
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You see?

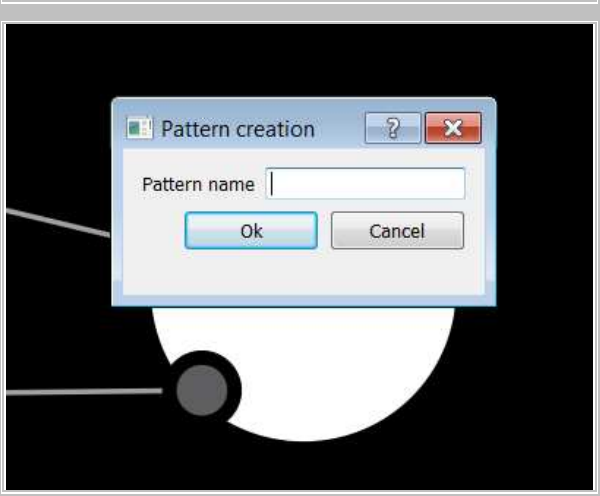


You see ?



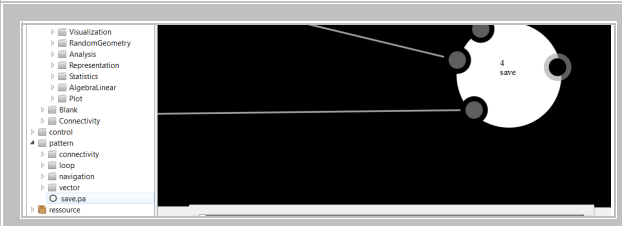
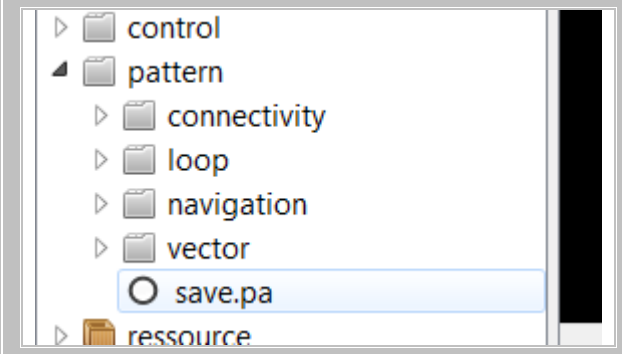
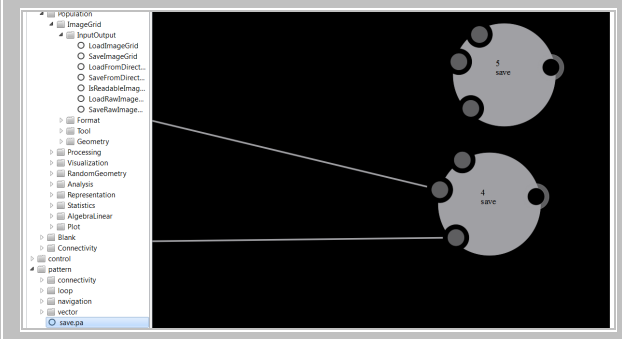
Now we think this save sub process can be interesting for other projects because it's a repetitive pattern.

Click left on it and **select** "create pattern".



Name your pattern "save" and **press** the "ok" button.

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	On the pattern directory of the glossary ...
	... you can now see a pattern named "save.pa".
	**Drag & drop** save.pa as you create an operator.
	And **create** some copy of your pattern.
The end.	In next tutorial, we see how to share your work.

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